

BERGEN FLESLAND, Norway

WMO: 013110

Lat: 60.293N

Lon: 5.218E

Elev: 170

StdP: 14.61

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	19.1	22.1	5.2	7.2	24.8	10.0	9.2	26.7	32.4	40.0	29.8	40.1	3.5	100	0.654

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	10.4	74.5	61.5	70.6	60.1	67.6	58.6	63.8	70.6	62.0	67.8	60.4	65.5	8.3	340

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
61.2	81.2	65.8	59.6	76.7	64.1	58.1	72.6	62.6	29.1	70.8	27.9	68.0	26.8	65.7	70.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation	n=5 years		n=10 years		n=20 years		n=50 years			
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(4) 27.4	23.7	20.8		16.2	80.0	3.7	4.2	13.6	83.0	11.4	85.5	9.3	87.8	6.7	90.9	
(5)			DB	14.7	66.3	3.9	2.1	11.9	67.8	9.6	69.1	7.4	70.3	4.6	71.8	(p)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	45.7	35.3	34.9	37.3	43.1	48.9	53.9	58.5	58.6	53.9	46.8	40.4	36.5
	DBStd	10.20	5.97	5.61	4.79	5.28	5.78	4.81	5.01	4.75	4.69	5.35	6.09	6.82
	HDD50	2440	456	422	395	216	89	12	1	0	15	124	290	420
	HDD65	7056	921	842	860	656	500	336	212	209	333	564	738	885
	CDD50	884	0	0	0	10	55	128	265	266	132	26	2	0
	CDD65	26	0	0	0	0	1	3	11	10	1	0	0	0
	CDH74	125	0	0	0	0	5	12	70	37	1	0	0	0
	CDH80	13	0	0	0	0	0	1	10	2	0	0	0	0
	WSAvg	8.7	10.1	9.4	9.2	8.4	8.5	8.3	7.3	7.6	8.1	8.9	8.8	9.6
	PrecAvg	79.8	6.9	5.4	5.9	4.2	3.9	4.9	5.4	6.8	10.2	9.4	9.0	8.4
(15) Precipitation	PrecMax	115.4	15.7	13.2	15.4	10.7	8.0	9.8	10.7	13.5	19.9	19.8	15.1	13.5
	PrecMin	56.5	1.7	0.2	0.8	0.4	1.1	0.2	1.0	0.2	2.4	2.0	3.1	1.4
	PrecStd	15.2	3.7	3.3	3.7	2.0	2.0	2.3	2.6	3.2	4.2	4.1	3.6	3.9
	PrecStd	15.2	3.7	3.3	3.7	2.0	2.0	2.3	2.6	3.2	4.2	4.1	3.6	3.9
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	48.0	47.8	52.6	64.7	73.4	75.6	81.3	78.8	71.1	63.7	54.9	50.7	
	0.4% MCWB	43.0	44.2	44.6	50.9	57.8	59.9	64.5	64.3	59.7	55.1	49.1	47.4	
	2% DB	46.2	45.6	47.4	59.3	67.7	70.1	75.0	73.3	66.5	58.8	52.0	48.5	
	2% MCWB	43.3	43.5	42.2	48.3	54.7	57.7	62.5	62.0	57.0	52.2	48.4	46.3	
	5% DB	44.8	44.4	45.2	55.2	63.0	65.7	70.3	69.5	63.8	56.0	50.3	47.0	
	5% MCWB	42.6	42.5	41.8	46.4	52.4	55.7	61.3	60.8	56.5	51.5	47.1	44.9	
	10% DB	43.2	42.9	43.9	51.5	58.9	62.2	66.8	66.3	61.0	54.0	48.6	45.6	
	10% MCWB	41.2	41.1	41.1	44.7	50.4	54.6	59.5	59.6	55.5	50.4	45.9	43.6	
	0.4% WB	45.8	45.7	45.9	53.1	59.8	61.6	66.4	66.1	62.0	56.7	51.6	48.9	
	0.4% MCDB	46.4	46.3	48.0	61.7	71.2	72.2	76.4	73.5	66.8	59.5	52.8	49.5	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	44.4	44.1	44.3	49.7	56.2	59.2	64.0	63.9	59.7	54.7	50.0	47.2	
	2% MCDB	45.2	44.9	45.8	55.7	64.3	67.0	71.8	69.9	63.9	57.0	51.4	47.9	
	5% WB	43.1	43.0	43.1	47.7	53.7	57.2	62.2	62.3	57.8	53.1	48.3	45.6	
	5% MCDB	44.0	43.9	44.7	52.9	61.3	63.3	68.3	67.4	61.7	55.4	49.7	46.6	
	10% WB	41.6	41.4	41.7	46.0	51.4	55.5	60.6	60.6	56.3	51.3	46.3	43.8	
	10% MCDB	42.9	42.5	43.3	49.9	57.6	60.5	65.7	65.0	59.9	53.5	48.0	45.1	
(35) Mean Daily Temperature Range	MDBR	7.2	8.0	9.0	11.1	11.9	11.1	10.5	10.4	9.7	8.9	7.7	7.2	
	5% DB	7.6	7.8	10.3	17.0	18.4	17.7	17.1	15.3	13.6	10.6	8.2	7.6	
	5% WB	7.0	6.7	7.0	8.9	8.9	8.4	7.6	7.2	7.2	6.7	7.0	7.1	
	MCDBR	7.3	7.1	8.5	13.6	16.8	14.8	14.6	13.2	10.6	8.8	7.7	7.3	
	MCWBR	7.3	6.9	6.9	7.8	8.9	8.1	7.3	7.0	7.0	7.0	7.9	7.4	
	MCWBR	7.3	6.9	6.9	7.8	8.9	8.1	7.3	7.0	7.0	7.0	7.9	7.4	
(40) Clear-Sky Solar Irradiance	taub	0.289	0.319	0.341	0.371	0.360	0.355	0.370	0.367	0.351	0.329	0.295	0.395	
	taud	2.148	2.275	2.332	2.321	2.429	2.475	2.456	2.462	2.494	2.434	2.232	2.609	
	Ebn at Noon	145	213	248	262	275	278	270	262	246	211	143	102	
	Edh at Noon	14	21	28	33	32	31	31	29	24	18	12	10	
(44) All-Sky Solar Radiation	RadAvg	72	239	580	1095	1570	1673	1481	1137	697	338	107	45	
	RadStd	11	43	88	109	152	200	141	112	84	49	15	6	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	-166	N/A	N/A	N/A
Regional (39 neighbors)	N/A	+1.22	N/A	N/A	N/A	N/A	-151	-221	N/A	N/A

Nomenclature: See separate page